

### A.5. Steering family-specific latents

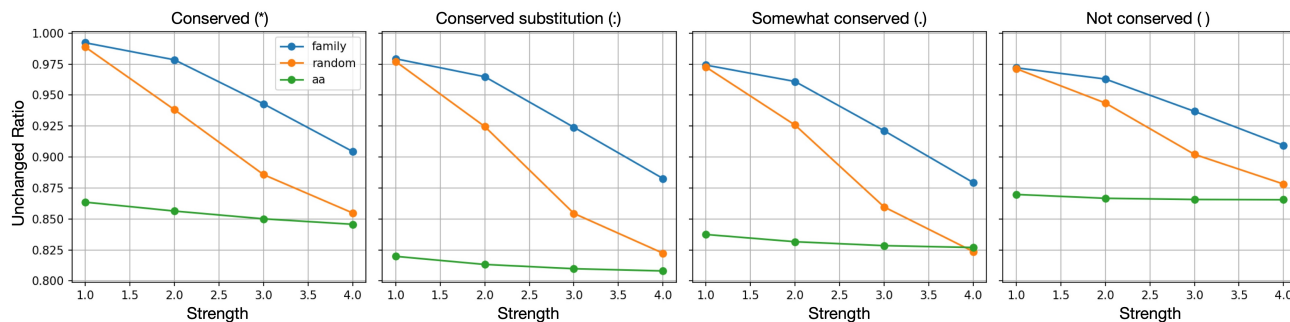


Figure 7. The fraction of residues that remains unchanged after steering, averaged for 2089 family specific features revealed using SAE.

We steer each family-specific latent and analyze the resulting sequence changes compared to steering random latents or latents corresponding to amino acid features. Steering refers to clamping of the activation of a specific latent to a fixed level—either increasing it to amplify the feature or decreasing it to suppress it. For each family-specific latent corresponding to an InterPro family, we identify the top-activating sequences, conduct protein-protein BLAST (Ye et al., 2006), and align sequences within the same InterPro family to obtain a multiple sequence alignment using Clustal Omega (Sievers et al., 2011). Finally, we determine the maximum activation of a family-specific latent and clamp the activation of the activating residues to  $1\times$ ,  $2\times$ ,  $3\times$ , and  $4\times$  the maximum value. As a baseline, we steer the residues which activated the family-specific latent with two random latent dimensions (1 and 11) and two amino acid specific latent dimensions (3267 for alanine and 3830 for glycine in the SAE with a hidden dimension of 4096 from layer 24). Because we choose two latents as representative random and amino acid latents, we take the average of the unchanged ratio. The unchanged ratio is the fraction of residues across the whole sequence that remains unchanged after steering.

Figure 7 compares the fraction of residues that remains unchanged after steering of family-specific (labeled “family”), random (labeled “random”), and amino acid (labeled “aa”) latents, by the degree of conservation (decreasing order, left to right). For any degree of conservation, steering a family-specific latent has the least effect in the decoded sequence for the same steering strength, suggesting that family-specific latents capture underlying evolutionary constraints that resist disruption. The difference in unchanged ratio between family-specific and random latents is larger in conserved residues, implying steering family-specific latents tends to keep the residues with high consensus. Still, it is interesting that absolute unchanged ratio is roughly in a similar range for any degree of conservation.

### A.6. Classifying latents by family specificity

We evaluate the ability of a single SAE latent to classify a protein as belonging to a specific InterPro-defined protein family (Paysan-Lafosse et al., 2023). To do this, we determine the optimal normalized activation threshold that maximizes the F1 score for binary classification of protein families. Specifically, we sweep threshold values from 0.1 to 0.9 in 0.1 increments and select the threshold that results in the highest F1 score. Our final evaluation is conducted on a held-out test set.

For our evaluation, we use protein sequences from Swiss-Prot (Boeckmann et al., 2003). To reduce sequence redundancy, we cluster all Swiss-Prot sequences with a length of 1022 residues or fewer at 30% sequence identity using MMseqs2 (`mmseqs easy-cluster swissprot.fasta session tmp --min-seq-id 0.3 -c 0.9 -s 8 --max-seqs 1000000 --cluster-mode 1`) (Steinegger & Söding, 2017). Given the low sequence identity between clustered sequences, we perform a random train-test split for classification evaluation.

To determine which protein family to assess for each latent, we identify the InterPro family of the sequence that exhibits the highest activation for that latent. The F1 score is then computed for this family using the thresholding approach described above.

## A.7. Classifying latents by activation pattern

Table 2. Activation pattern classification criteria

| Category              | Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dead Latent           | If the latent is never activated by any test sequences, it is classified a dead latent.                                                                                                                                                                                                                                                                                                                                                              |
| Not Enough Data       | If less than 5 sequences activate the latent then we say there is not enough data.                                                                                                                                                                                                                                                                                                                                                                   |
| Periodic              | Features that exhibit consistent activation patterns at regular intervals. These features must satisfy: (1) a high frequency of activation at specific positions (over 50% of distances between activations are the same two values), (2) a large number of activation regions (there are more than 10 activations per sequence), and (3) relatively short contiguous activation spans (median length of the top activating contig is less than 10). |
| Point                 | Features that activate in a highly localized manner, defined by a single, prominent activation site (the median length of the highest activating region is 1).                                                                                                                                                                                                                                                                                       |
| Motif (Short: 1-20)   | Features that activate in short contiguous regions (median length of the highest activating region is $> 1$ and $< 20$ ) and have an overall mean activation coverage of less than 80%.                                                                                                                                                                                                                                                              |
| Motif (Medium: 20-50) | Features that activate in short contiguous regions (median length of the highest activating region is $\geq 20$ and $< 50$ ) and have an overall mean activation coverage of less than 80%.                                                                                                                                                                                                                                                          |
| Motif (Long: 50-300)  | Features that activate in short contiguous regions (median length of the highest activating region is $\geq 50$ and $< 300$ ) and have an overall mean activation coverage of less than 80%.                                                                                                                                                                                                                                                         |
| Whole                 | Features that are active across nearly the entire sequence (overall mean activation coverage of greater than 80%).                                                                                                                                                                                                                                                                                                                                   |
| Other                 | Features that do not meet any of the above criteria are classified as "other."                                                                                                                                                                                                                                                                                                                                                                       |

### A.8. Testing SAE hyperparameters

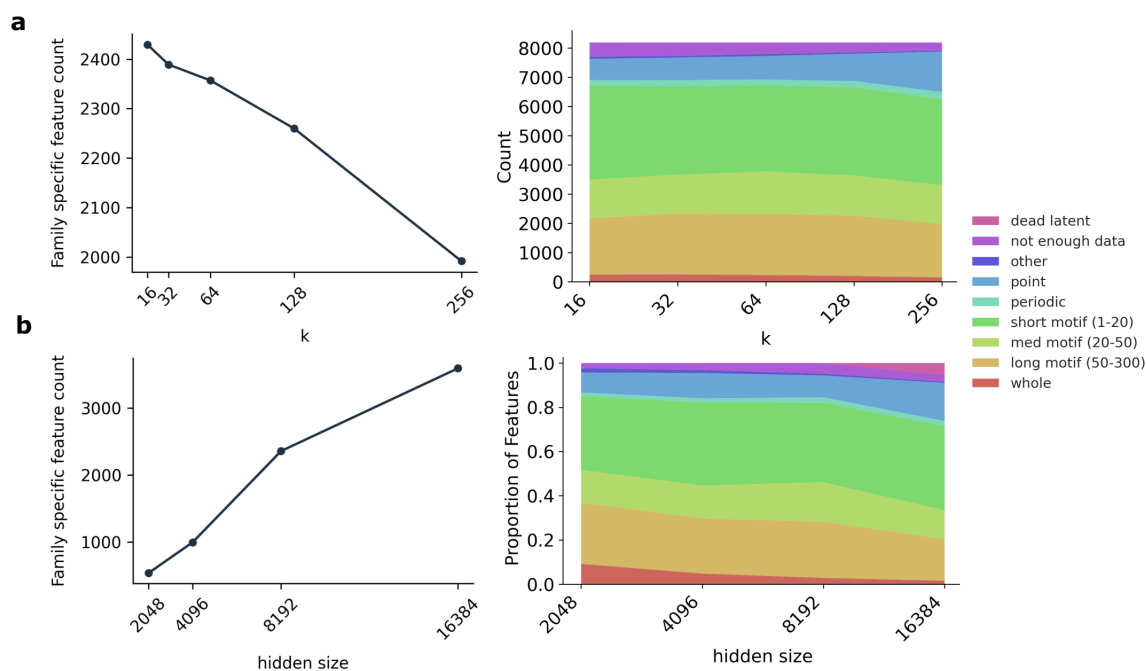


Figure 8. Testing SAE hyperparameters. **a.**  $k$  parameter sweep (other hyperparameters held constant, layer=24, hidden size=8096) showing the number of family specific features (left) and the features categorized by activation pattern (right). **b.** SAE hidden size hyperparameter sweep (other hyperparameters held constant, layer=24,  $k$ =64) showing the number of family specific features (left) and the features categorized by activation pattern (right).